INDIAN INSTITUTE OF INFORMATION TECHNOLOGY ALLAHABAD

Department of Information Technology

IMAGE & VIDEO PROCESSING

PROJECT DESIGN REPORT

for

Multi-Feature Fusion Based Cross-Camera Person Tracking System

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Base Research Paper

Cross-Camera Tracking Model and Method Based on Multi-Feature Fusion

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1. Project Overview

Cross-camera person tracking is an important problem in modern video surveillance and intelligent vision systems. A person observed by one camera may later appear in another camera with a different viewpoint, illumination, scale, background, or partial occlusion. These variations make it difficult to maintain a consistent identity across multiple camera views.

The proposed project, titled **Multi-Feature Fusion Based Cross-Camera Person Tracking System**, aims to design a computer vision pipeline capable of detecting and tracking persons within individual cameras and subsequently associating the same person across multiple cameras.

The proposed approach combines appearance information with motion, trajectory, temporal, and other contextual features to produce a more robust representation for cross-camera matching.

2. Project Scope

The scope of the proposed project includes the design and development of a system capable of processing multiple video streams and maintaining consistent person identities across different camera views.

The major activities included in the project are:

- Acquisition of multiple video streams.
- Frame extraction and preprocessing.
- Person detection.
- Single-camera multi-person tracking.
- Local identity assignment.
- Appearance feature extraction.
- Motion and trajectory analysis.
- Multi-feature fusion.
- Cross-camera person association.
- Global identity assignment.
- Visualization of tracking results.
- Performance evaluation and analysis.

The project focuses primarily on person tracking in surveillance-like environments and does not attempt to address unrelated object categories.

3. Project Objectives

The primary objective of this project is to design and implement a robust system for tracking persons across multiple camera views using a combination of complementary visual and contextual features.

The specific objectives are:

1. To process video streams from multiple cameras.
2. To detect persons in individual video frames.
3. To track persons within each camera and maintain local identities.
4. To extract discriminative appearance features for each person.
5. To analyze motion and trajectory information.
6. To combine multiple features using a feature fusion strategy.
7. To compare observations of persons appearing in different cameras.
8. To associate observations belonging to the same real-world person.
9. To assign and maintain a consistent global identity.
10. To visualize and evaluate the tracking results.

4. Base Research Paper

The proposed system is conceptually based on the following research paper:

> P. Zhang, S. Wang, W. Zhang, W. Lei, X. Zhao, Q. Jing, and M. Liu, "Cross-Camera Tracking Model and Method Based on Multi-Feature Fusion," *Symmetry*, vol. 15, no. 12, Article 2145, 2023.

DOI: https://doi.org/10.3390/sym15122145

The base paper addresses the challenge of tracking objects across different camera views by utilizing multiple complementary features. The central idea motivating this project is that relying on a single feature may not be sufficient when large appearance or viewpoint variations occur.

The proposed project adapts this idea to person tracking by combining:

- Appearance features.
- Motion information.
- Trajectory information.
- Temporal information.

These features are used together to improve the reliability of cross-camera person association.

5. Proposed System Design

The proposed system follows a modular pipeline architecture. Each module performs a specific operation and passes its output to the next stage.

The overall processing pipeline is:

Video Streams

↓

Frame Processing

↓

Person Detection

↓

Single-Camera Tracking

↓

Feature Extraction

↓

Multi-Feature Fusion

↓

Cross-Camera Association

↓

Global Identity Assignment

↓

Visualization and Storage

The modular structure allows individual components to be improved or replaced without redesigning the complete system.

6. System Architecture

The proposed architecture consists of three major layers:

1. Input and Perception Layer
2. Feature and Association Layer
3. Identity and Output Layer

6.1 Input and Perception Layer

This layer receives video streams from multiple cameras and performs initial processing operations such as frame extraction, preprocessing, person detection, and single-camera tracking.

6.2 Feature and Association Layer

This layer extracts appearance and contextual features from tracked persons. The extracted features are combined using a feature fusion strategy and subsequently used for cross-camera similarity computation.

6.3 Identity and Output Layer

This layer assigns global identities based on association results and presents the final tracking information through visualization and storage modules.

7. System Architecture Diagram

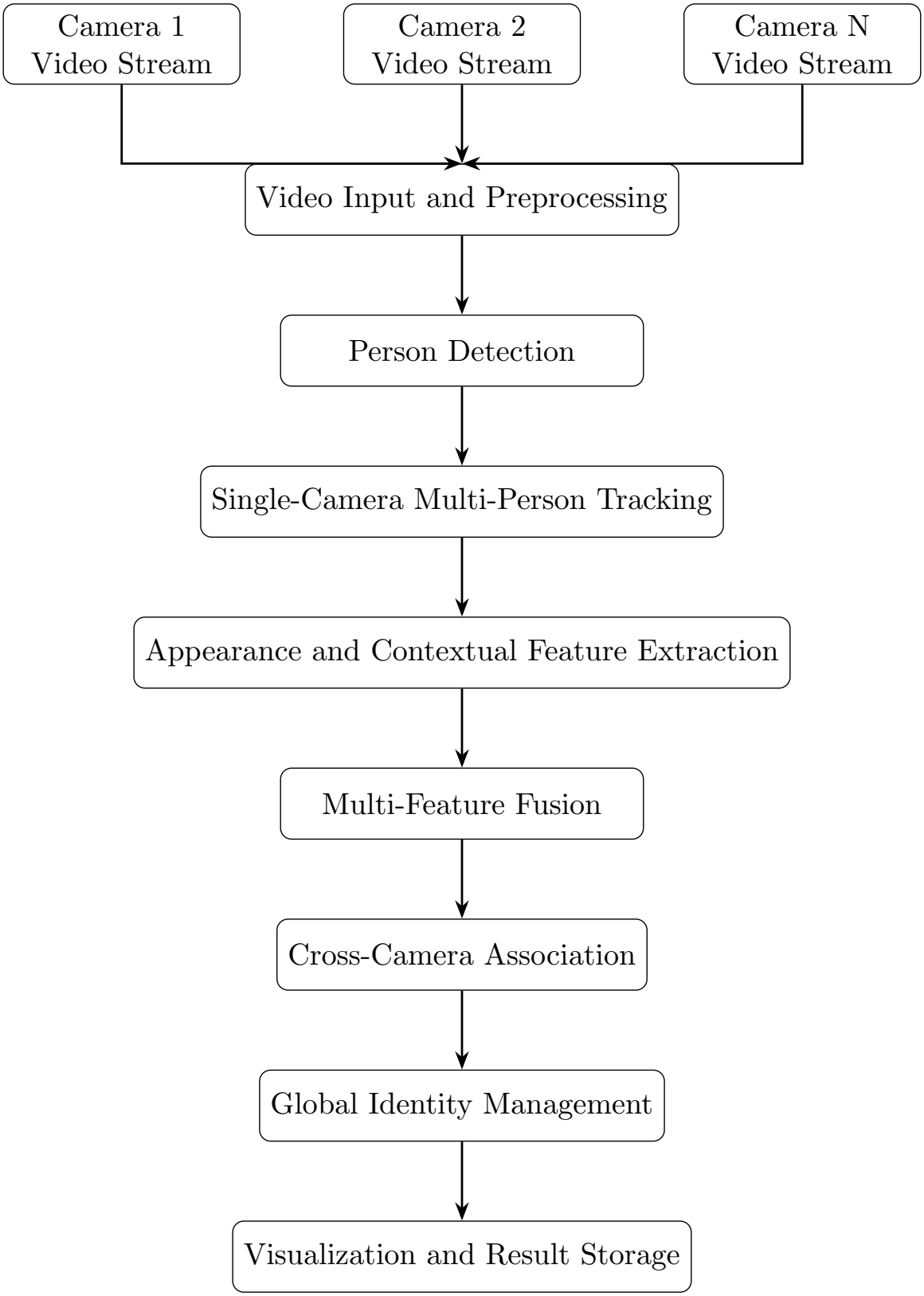


Figure 1: Overall System Architecture

8. Detailed Processing Block Diagram

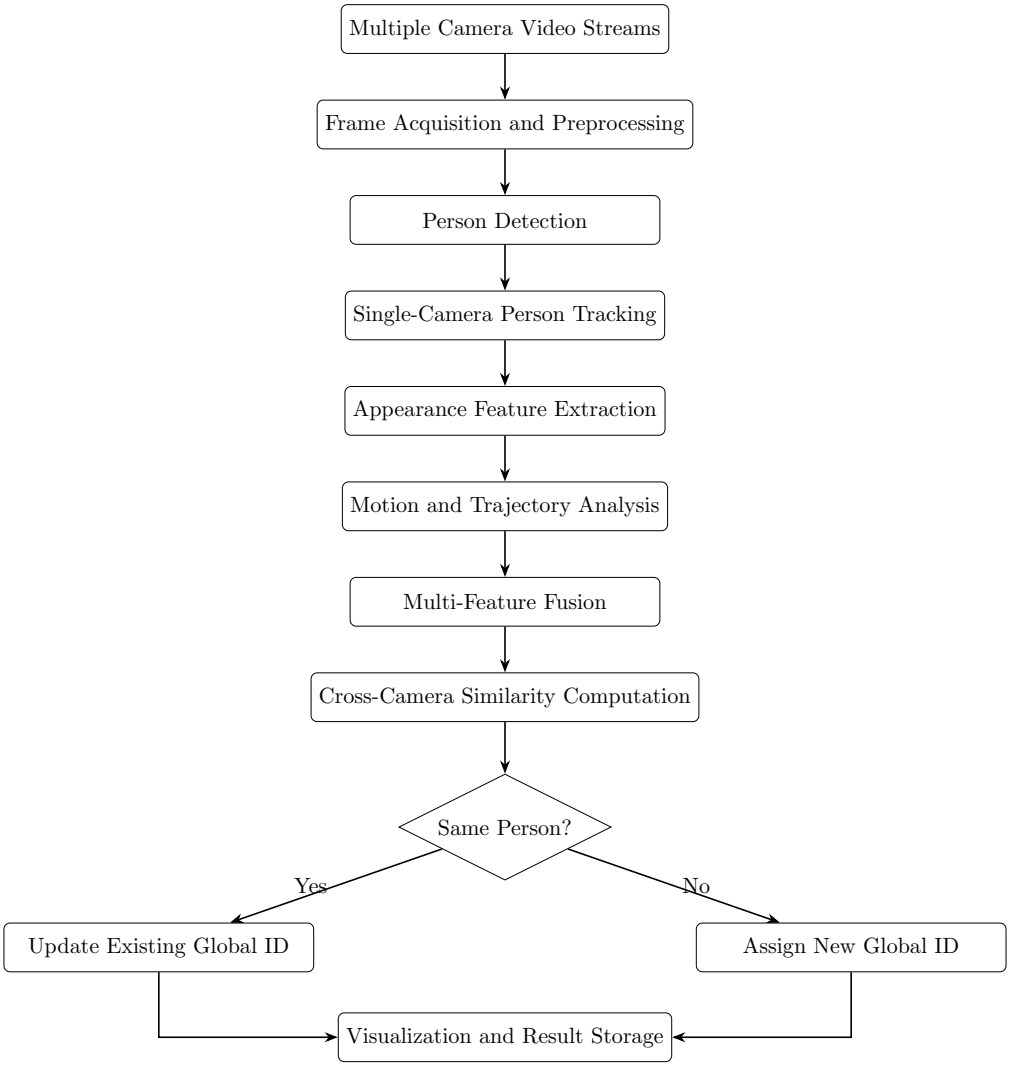


Figure 2: Detailed Processing Block Diagram

9. Detailed Module Design

9.1 Video Input Module

The Video Input Module receives video streams from one or more cameras. Each video stream is processed independently during the initial stages.

The module performs:

- Video loading.
- Frame extraction.
- Resolution normalization.
- Basic preprocessing.

9.2 Person Detection Module

The Person Detection Module identifies person instances within each frame.

For every detected person, the output contains:

$$D_i = (x_i, y_i, w_i, h_i, c_i)$$

where:

- (x_i, y_i) represents the bounding box position.
- w_i and h_i represent width and height.
- c_i represents the detection confidence.

9.3 Single-Camera Tracking Module

The tracking module associates person detections across consecutive frames.

Each tracked object receives a local identity:

$$T_i = \{\text{Local ID}, \text{Bounding Box}, \text{Trajectory}\}$$

This module helps maintain identity continuity within a single camera.

9.4 Appearance Feature Extraction Module

For each tracked person, visual appearance features are extracted.

These features may capture:

- Clothing appearance.
- Color information.
- Visual texture.
- Deep visual embeddings.

The appearance representation is denoted as:

$$F_a$$

9.5 Motion and Trajectory Analysis Module

The motion module analyzes the movement of tracked persons.

The trajectory representation may include:

$$F_m = \{\Delta x, \Delta y, v_x, v_y, t\}$$

where position changes, velocity, and temporal information contribute to cross-camera association.

9.6 Multi-Feature Fusion Module

The Multi-Feature Fusion Module combines complementary features into a single representation.

A conceptual fusion model is:

$$F = w_a F_a + w_m F_m + w_t F_t$$

where:

- F_a represents appearance features.
- F_m represents motion features.
- F_t represents temporal features.
- w_a, w_m, and w_t are configurable weights.

The weights may be adjusted experimentally to determine their influence on cross-camera association performance.

9.7 Cross-Camera Association Module

This module compares fused feature representations from different cameras.

A conceptual similarity score is:

$$S(i,j) = \alpha S_a(i,j) + \beta S_m(i,j) + \gamma S_t(i,j)$$

where:

- S_a represents appearance similarity.
- S_m represents motion similarity.
- S_t represents temporal similarity.

If:

$$S(i,j) > \tau$$

the observations may be associated with the same global identity.

9.8 Global Identity Management Module

The Global Identity Management Module maintains a mapping between local camera-specific identities and global identities.

For example:

$$G_k = \{C_1 : T_4, C_2 : T_7, C_3 : T_2\}$$

indicates that the specified local tracks from different cameras correspond to the same person.

9.9 Visualization Module

The visualization module displays:

- Camera streams.
- Person bounding boxes.
- Local identities.
- Global identities.
- Trajectories.
- Association information.

10. Data Flow Design

The data flow through the system can be summarized as follows:

1. Multiple cameras generate video streams.
2. Frames are extracted and preprocessed.
3. Person detection identifies person instances.
4. The tracking module creates local trajectories.
5. Features are extracted for every tracked person.
6. Multiple features are combined into fused representations.
7. Candidate persons from different cameras are compared.
8. Similarity scores are calculated.
9. A global identity is assigned or updated.
10. Results are visualized and stored.

11. Feature Fusion Design

A single feature can become unreliable under difficult conditions. For example, appearance features may change due to illumination, while trajectory information may become unreliable when the camera viewpoints are unrelated.

The proposed system therefore combines multiple features.

The feature fusion process can be represented as:

$$F_{\text{fusion}} = w_1 F_{\text{appearance}} + w_2 F_{\text{motion}} + w_3 F_{\text{trajectory}} + w_4 F_{\text{temporal}}$$

The purpose of feature fusion is to increase the robustness of person association by using complementary information.

12. System Flowchart

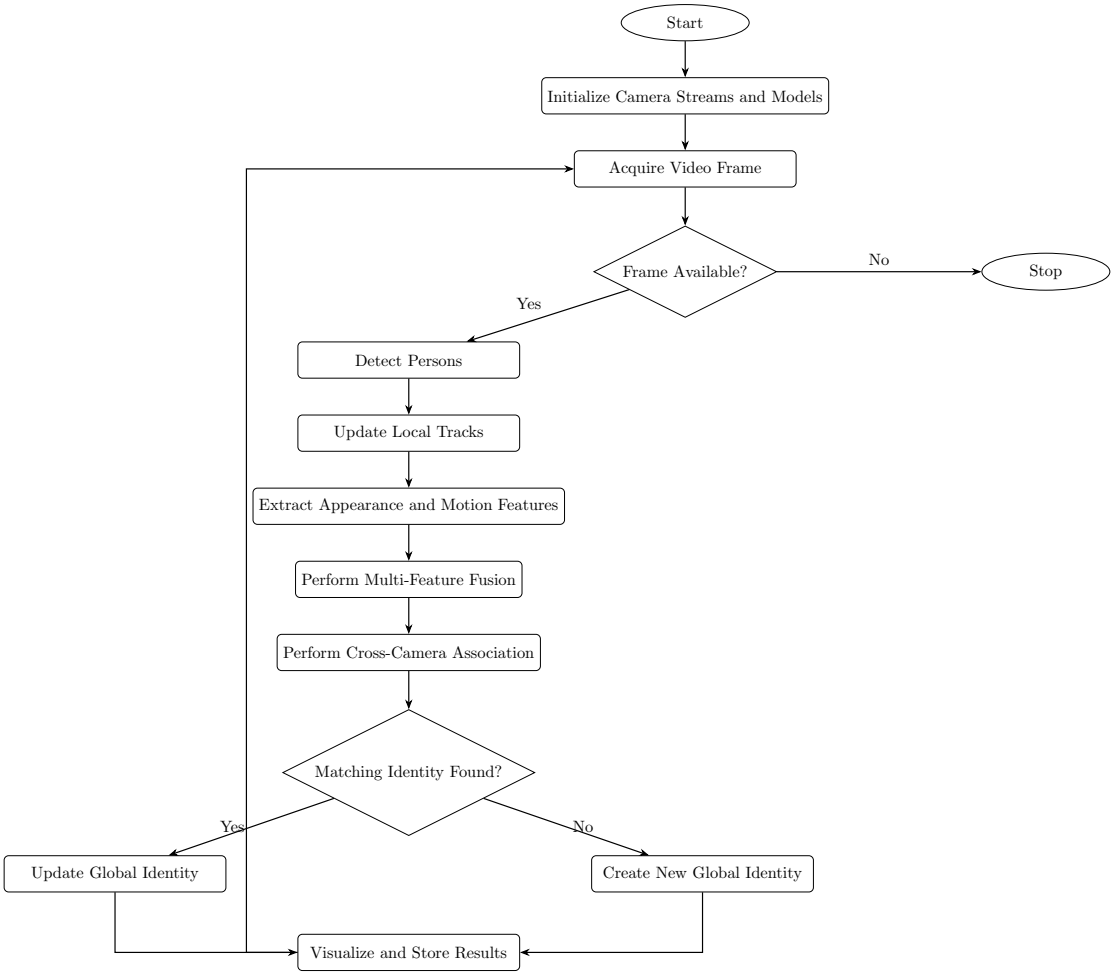


Figure 3: Operational Flowchart of the Proposed System

13. Implementation Design

The proposed implementation will use a modular software structure.

13.1 Technology Stack

The expected technology stack is:

- **Programming Language:** Python
- **Video Processing:** OpenCV
- **Deep Learning:** PyTorch
- **Numerical Processing:** NumPy
- **Visualization:** OpenCV and Matplotlib

13.2 Proposed Software Modules

The implementation can be organized into the following modules:

- `video_input.py`
- `detector.py`
- `tracker.py`
- `feature_extractor.py`
- `feature_fusion.py`
- `cross_camera_association.py`
- `global_identity.py`
- `visualization.py`
- `main.py`

This separation improves maintainability and allows independent experimentation with individual components.

14. Evaluation Plan

The proposed system will be evaluated using multiple criteria.

14.1 Detection Performance

Detection quality can be evaluated using precision and recall.

14.2 Tracking Performance

Tracking quality can be evaluated using:

- IDF1.
- Identity Precision.
- Identity Recall.
- Number of Identity Switches.

14.3 Cross-Camera Association Performance

The ability of the system to correctly associate the same person across different cameras will be evaluated by comparing predicted global identities with available ground truth data.

14.4 Computational Performance

The system will also be evaluated based on:

- Processing time.
- Frames per second.
- Memory usage.
- Scalability with multiple cameras.

15. Current Limitations

The proposed system may face challenges under the following conditions:

- Severe occlusion.
- Poor illumination.
- Large viewpoint changes.
- Similar clothing among multiple persons.
- Low-resolution video.
- Long time gaps between camera observations.

These limitations can affect both single-camera tracking and cross-camera association.

16. Future Enhancements

Possible future improvements include:

- Adaptive feature weighting.
- Advanced deep person re-identification models.
- Automatic camera topology learning.
- Real-time deployment.
- Large-scale multi-camera support.
- Camera calibration and spatial transformation.
- Edge-based implementation.
- Advanced tracking analytics.

17. Conclusion

This Design Report presents the proposed design of a **Multi-Feature Fusion Based Cross-Camera Person Tracking System** for the Image & Video Processing course project.

The design is motivated by the limitations of relying solely on appearance information when associating persons across different camera views. The proposed architecture combines person detection, single-camera tracking, appearance analysis, motion and trajectory analysis, multi-feature fusion, cross-camera association, and global identity management.

The modular architecture, block diagram, and flowchart provide a structured blueprint for the future implementation and evaluation of the proposed system.

References

[1] P. Zhang, S. Wang, W. Zhang, W. Lei, X. Zhao, Q. Jing, and M. Liu, "Cross-Camera Tracking Model and Method Based on Multi-Feature Fusion," *Symmetry*, vol. 15, no. 12, Article 2145, 2023. DOI: https://doi.org/10.3390/sym15122145.